

Anomaly Detection of Deepfake Audio Based on Real Audio Using Generative Adversarial Network Model

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Abstract

Unseen zero-day deepfake audio attacks often bypass supervised binary neural networks. This work proposes an unsupervised anomaly detection framework trained exclusively on authentic real audio signals using an Autoencoder-GAN (AE-GAN). Any synthetic audio or tampered watermark exhibits high reconstruction error scores, flagging it as fake.

Key Methodology & Architecture

Autoencoder-GAN (AE-GAN), Reconstruction Error Loss, Latent Space Mapping, Anomaly Thresholding.

Datasets & Benchmark Environments

VoxCeleb1, AudioSet Pristine Speech Subset, Zero-Day TTS Generators.

Performance Metrics & Graph Axis Parameters

Reported Results: Zero-Day Deepfake Detection AUC = 0.945, False Positive Rate (FPR) = 2.4%, Precision = 96.1%.

Graph Parameter Mapping: X-axis: Anomaly Reconstruction Threshold; Y-axis: True Positive Rate vs False Positive Rate (ROC Curve).

Comparative Analysis

Advantages (Pros)	Limitations (Cons)
Does not require deepfake training samples; highly robust to unknown zero-day voice clones.	Slightly lower accuracy on subtle, highly pristine generative audio.

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